

SAYAK CHOWDHURY

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EDUCATION

International Institute of Information Technology Bangalore

CGPA: 3.33/4

Master of Science (by Research) in Computer Science

Aug. 2023 – Jul. 2026

Coursework: Maths for Machine Learning, Machine Learning, Software Testing, Visual Recognition

Maulana Abul Kalam Azad University of Technology, West Bengal

CGPA: 8.79/10

B.Tech in Applied Electronics and Instrumentation Engineering

Aug. 2018 – Jun. 2022

Coursework: Artificial Intelligence, Control Theory, Big Data Analysis, DBMS

EXPERIENCE

Nykaa

Jan. 2026 – Present

Data Science Intern

Bangalore, India

- Executing experimentation for emerging initiatives to prototype ideas and support data-driven decision making.
- Designing and evaluating personalized recommendation models by experimenting with user and product embedding techniques and diverse model architectures across the Nykaa ecosystem.

Siemens Technology and Services

Aug. 2024 – Nov. 2025

Research Intern

Bangalore, India

- Engineered a method for testing of systems composed of multiple deep learning models as measured by evaluating pipelines with **3+ object detectors** and developing a novel testing framework modelling system behaviour.
- Co-developed one of only **3 existing formal verification techniques** for object detection models, culminating in a peer-reviewed ICANN 2025 publication on set-based reachability analysis.

Tata Consultancy Services Ltd.

Jul. 2022 – Aug. 2023

Assistant System Engineer

Kolkata, India

- Accelerated enterprise-grade Oracle database migration at Pfizer by reducing task duration from 16 hours to 3 hours (**81% time savings**), by developing automation scripts for client-side CRUD workflows and bulk data fixes.
- Developed and maintained over 30 scripts for seamless deployment and migration processes in production environments, optimizing client-side CRUD operations on large-scale Oracle databases while decreasing execution time by an average of 25%.

PUBLICATIONS (* EQUAL CONTRIBUTION)

1. H.* Khandelwal, Chowdhury, S.*, and M. D'Souza. **Robustness Verification for Object Detectors Using Set-Based Reachability Analysis**. In *Proceedings of the 34th International Conference on Artificial Neural Networks (ICANN 2025)*. Springer, 2025
2. Chowdhury, S. and M. D'Souza. **SETA: Statistical Fault Attribution for Compound AI Systems**. In *IEEE/ACM 48th International Conference on Software Engineering (ICSE 2026) Co-located Conferences*. CAIN, 2026

PROJECTS

ASR Spell Correction for Medical Transcripts | *Transformers, T5, BERT*

July 2025 – August. 2025

- Developed a specialized correction system targeting ASR errors in medical conversations by combining NER + POS tagging for noun extraction and building a curated error-pattern database of 500+ medication names and terms.
- Benchmarked traditional baselines (Levenshtein edit distance, n-gram LM, dictionary-based correction) against fine-tuned transformer models (BERT for contextual masking, T5 seq2seq correction), improving noun-specific accuracy by 18% and BLEU score by 12%.
- Delivered a domain-adaptive spell correction pipeline that reduces transcription risk in medical ASR outputs, enabling safer downstream tasks such as clinical note generation and decision support systems.

Text-based RAG from scratch | *Python, huggingface, Ollama*

Feb. 2025 – Mar. 2025

- Implemented a local retrieval-augmented chatbot using Ollama embeddings and a Llama-based instruct model, with custom cosine similarity vector score for context retrieval and streaming chat responses in under 100 lines.

Benchmarking Bias in Multilingual LLMs <i>Multilingual LLMs, Responsible AI</i>	May 2024 – Jun. 2024
- Audited bias in 5 multilingual LLMs as measured by performance disparities on 500+ Indian-context sentence pairs, by benchmarking them using the CrowS-Pairs, IndiBias datasets and synthetic machine translated data. - Augmented existing datasets with human-generated translations to create richer benchmarks for empirical evaluation of fairness.	
Visual Question Answering using Transformer Models <i>ViT, BERT, LoRA</i>	Jan. 2024 – Mar. 2024
- Built a late-fusion multimodal transformer combining ViT image embeddings with BERT textual features. - Achieved 78% accuracy on MSCOCO VQA, with 30% fewer trainable parameters compared to full fine-tuning. - Fine-tuned model utilizing LoRA techniques from the PEFT library to demonstrate efficient multimodal adaptation of large transformers for VQA.	
End-to-End Microservices Testing <i>Postman, Docker Compose, Podman</i>	Oct. 2023 – Nov. 2023
Executed end-to-end testing across 5 most common e-commerce microservice scenarios as measured by identifying 1 critical usability bug, by using Postman and containerized environments with Docker Compose and Podman.	
Web Scraping for Amazon products <i>BeautifulSoup, Requests, Pandas</i>	Feb. 2023 – Mar. 2023
Extracted and structured 5,000 multilingual product reviews across 60+ Amazon listings, as measured by dataset scale and language diversity.	

CERTIFICATIONS

ML Summer School – Cohere Labs (Remote)	Jun. 2025 – Jul. 2025
Cohere for AI Open Science Community (Remote)	Dec. 2024 – present
ACM Bootcamp on AI Development Lifecycle – IIIT Bangalore	May. 2025
ACM Summer School on Responsible and Safe AI – IIT Madras	Jun. 2024
Present and Future Computing Systems Workshop – IISc Bangalore	Jan. 2024
Google Developer Group Kolkata	Aug. 2024 – present
Machine Learning Specialization (Stanford via Coursera)	Jun. 2023
Python for Everybody Specialization (University of Michigan via Coursera)	Jan. 2021
Google IT Automation Professional Certificate (Google via Coursera)	Nov. 2020
Data Analysis with Python: Zero to Pandas (Jovian)	Oct. 2020
AI For Everyone (DeepLearning.ai via Coursera)	Sep. 2020

POSITIONS OF RESPONSIBILITY / LEADERSHIP

RISE Organizing Committee, IIIT Bangalore	Dec. 2024 – Feb. 2025
<i>Senior Committee Member, Student Representative, Community Outreach, Speaker Invites</i>	
ADCOM 2024 Organizing Team, IIIT Bangalore	Nov. 2024 – Dec. 2024
<i>Organizing Committee Member, End-to-end logistics, Session Coordination</i>	
ICVGIP 2024 Local Team, IIIT Bangalore	Oct. 2024 – Dec. 2024
<i>Organizing Committee Member, Session lead, Author Coordination, Industry Engagement</i>	
ISEC 2024 Local Team, IIIT Bangalore	Jan. 2024 – Feb. 2024
<i>Organizing Committee Member, Speaker coordination, Scheduling, Attendee support</i>	
Samvaad Research Group, IIIT Bangalore	Jan. 2024 – present
<i>Organizer, Volunteer, Research talks, Q&A facilitation, Student engagement</i>	
IIIT-B Alumni Committee	Sep. 2023 – present
<i>Senior Committee Member, Alumni-student Outreach, Talks & Fundraising, Logistics, Database</i>	
Astronomy Club, IIIT Bangalore	Oct. 2023 – present
<i>Founding Member, Organizer, Outreach and Core Committee, Inter-college Outreach</i>	
Rotaract Club, RCCIIT Kolkata	Nov. 2018 – Dec. 2020
<i>Organizer, Outreach initiatives, Core Committee, Community service, Student-led programs</i>	

TECHNICAL SKILLS

Languages: Python, C, C++, SQL (SQLite), HTML/CSS, Lua, Powershell, Bash, awk, Markdown, $\text{\LaTeX}$, Java, CUDA
Developer Tools: Git, GitHub, Docker, Google Cloud Platform, VS Code, MongoDB, Postman, Vim, Nano, n8n
Libraries: Pandas, NumPy, Matplotlib, Seaborn, PyTorch, Tensorflow, Keras, Scikit-Learn, ONNX, Opencv, ollama